NovaWireless Transcript Analysis

NLP Feature Engineering for Call Outcome Prediction

Profanity Signals, Scenario-Distinctive Terms, and Target-Specific Lift Analysis

March 2026

Dataset: 60,000 synthetic calls • Corrected ledger • 12 months

Abstract

This paper presents the transcript-level NLP analysis conducted on the NovaWireless synthetic call dataset. We examine three dimensions of transcript content: profanity rates by outcome target, scenario-distinctive term distributions (clean vs. gaming vs. fraud), and target-specific term lift for three key outcome variables (bandaid_credit, proxy_vs_true_gap, and repeat_contact_30d). The analysis reveals that profanity rates are consistently higher in positive-outcome calls across all three targets, that scenario types produce highly distinctive term signatures, and that each outcome target has a unique linguistic fingerprint that could serve as a predictive feature for governance and monitoring systems.

1. Profanity Rates by Outcome Target

Figure 1 shows profanity rates segmented by outcome target direction. Across all three targets (proxy_vs_true_gap, bandaid_credit, and repeat_contact_30d), calls in the positive (target=1) class consistently show higher profanity rates than calls in the negative (target=0) class. For proxy_vs_true_gap, positive calls show approximately 53% profanity vs. 29% for negative. For bandaid_credit, the split is approximately 56% vs. 39%. For repeat_contact_30d, approximately 53% vs. 37%.

This finding has two implications. First, profanity is a viable signal for outcome prediction—calls with profanity are more likely to involve proxy-truth gaps, bandaid credits, and repeat contacts. Second, the relationship is not trivially explained by customer anger alone: the positive-class profanity rate exceeds 50% across all targets, suggesting that the majority of 'problem' calls contain some profanity in the transcript. This makes profanity presence a useful but blunt feature—high recall but limited precision as a standalone predictor.

Figure 1 — Profanity Rate by Outcome Target. Positive-outcome calls consistently show higher profanity rates.

2. Scenario-Distinctive Terms

Figure 2 presents a three-panel view of distinctive terms by scenario type (clean, gaming, and fraud), measured by log2 lift. Each panel shows the top 20 bigrams most uniquely associated with that scenario category relative to the full corpus.

Clean calls are dominated by positive service language: 'record great', 'account typically', 'days great', 'days hope'—terms reflecting successful interactions where the representative and customer reach a mutually satisfactory outcome. The IMEI term appearing in the clean panel reflects the legitimate IMEI verification process in standard calls.

Gaming calls show a distinctive pattern of emotional volatility mixed with resolution language: 'days good', 'happy shit', 'happy somebody', 'happy mess'—terms suggesting forced positivity layered over underlying frustration. The co-occurrence of 'happy' with profanity or negative nouns is a fingerprint of gamed interactions where the representative has manipulated the call toward an artificial resolution.

Fraud calls cluster around documentation and complaint language: 'account documented', 'account store', 'seriously hear', 'fixed hear', 'lied'—terms reflecting customers who are citing prior interactions, demanding accountability, or accusing the company of deception. These terms suggest multi-contact journeys where the customer has escalated beyond initial complaint into active documentation of perceived wrongs.

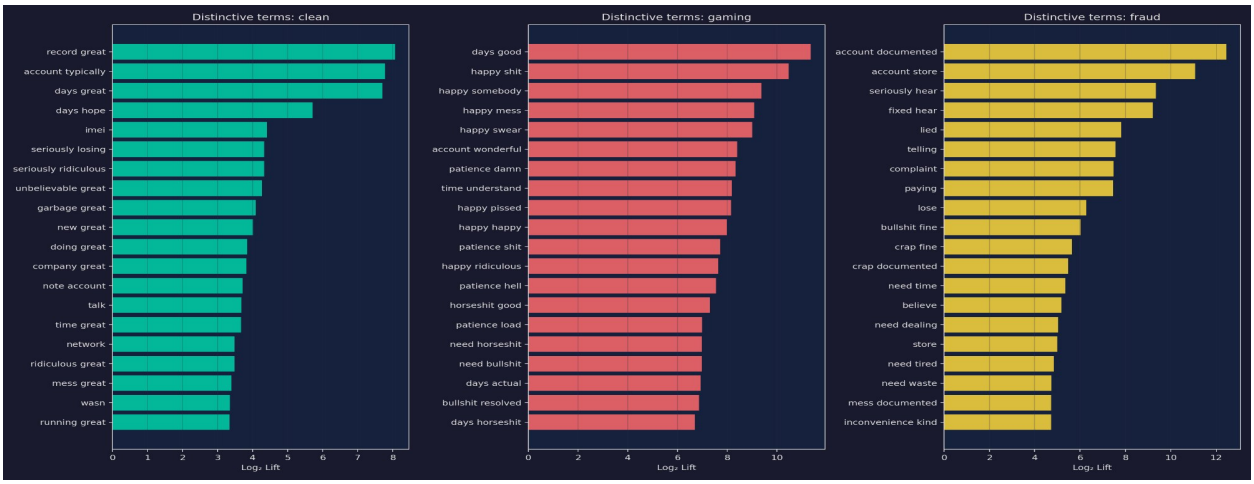

Figure 2 — Distinctive Terms by Scenario Type. Clean (left), Gaming (center), and Fraud (right) show distinct linguistic signatures.

3. Target-Specific Term Lift: Bandaid Credit

Figure 3 shows the top terms by log2 lift for the bandaid_credit target. The highest-lift terms are 'happy shit' and 'happy somebody' (both near 10.0 log2 lift), followed by 'happy mess', 'happy swear', and 'account wonderful'. The pattern is consistent: bandaid credits are most strongly associated with transcript language that mixes positive resolution framing ('happy', 'account wonderful', 'fine') with underlying distress signals ('shit', 'pissed', 'hell', 'damn'). This suggests that bandaid credits are applied as a resolution mechanism when the interaction has gone poorly but the representative needs to 'close' the call with an apparent positive outcome.

The middle portion of the lift chart shows terms like 'agree sick', 'agree wasted', 'fine pisses', and 'fine hell'—all combinations of agreement or acceptance language with negative affect. This linguistic pattern maps directly to the bandaid mechanism: the customer has been given something (a credit) and verbally agrees to close the interaction, but the underlying issue remains unresolved and the emotional tone betrays dissatisfaction that will resurface as a repeat contact.

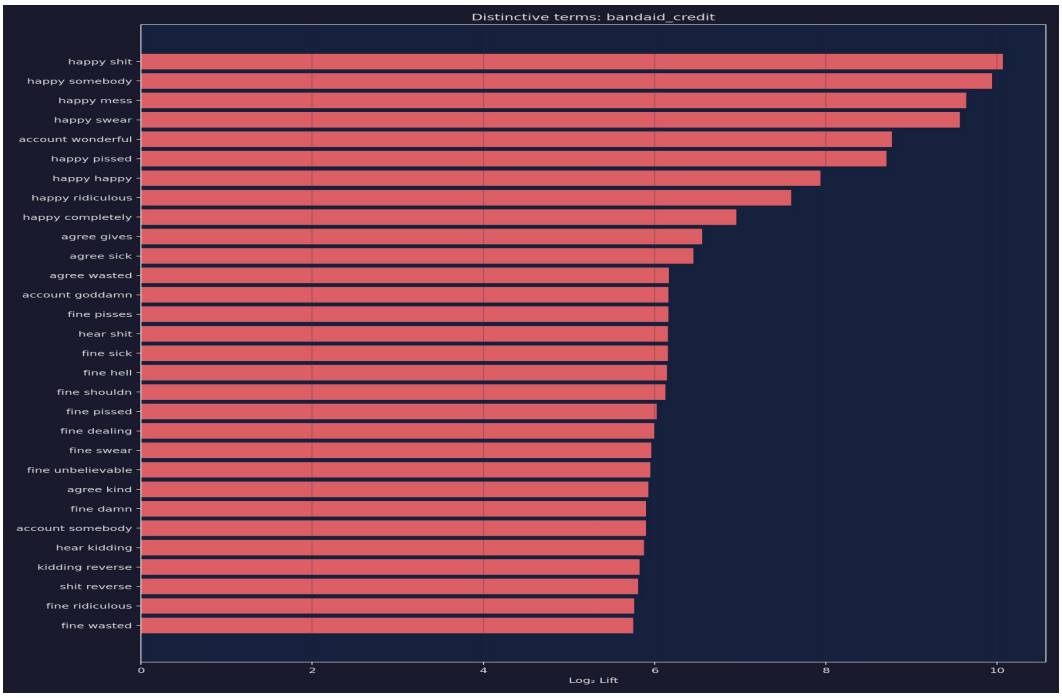

Figure 3 — Distinctive Terms: bandaid_credit. Forced positivity + profanity = bandaid signature.

4. Target-Specific Term Lift: Proxy vs. True Gap

Figure 4 shows terms most associated with calls where the proxy resolution flag diverges from true resolution. Top terms include 'ridiculous unfortunately', 'account tired', 'hours single', and 'account waste'—language expressing extended frustration with the account experience. The pattern differs from bandaid_credit terms: instead of forced positivity, proxy-gap calls feature sustained negative affect ('tired', 'waste', 'unfortunately', 'confused') combined with account-specific references ('account tired', 'account waste').

The mid-range terms include 'patience tired', 'doing confused', 'store unbelievable', and 'documented pisses'—terms suggesting customers who have been through multiple interactions and are documenting their frustration. The presence of 'kidding pay', 'kidding make', and 'line kidding' suggests expressions of disbelief at the situation. This linguistic profile describes a customer who has been told the issue is resolved (generating the proxy flag) but whose experience tells them otherwise (preventing true resolution).

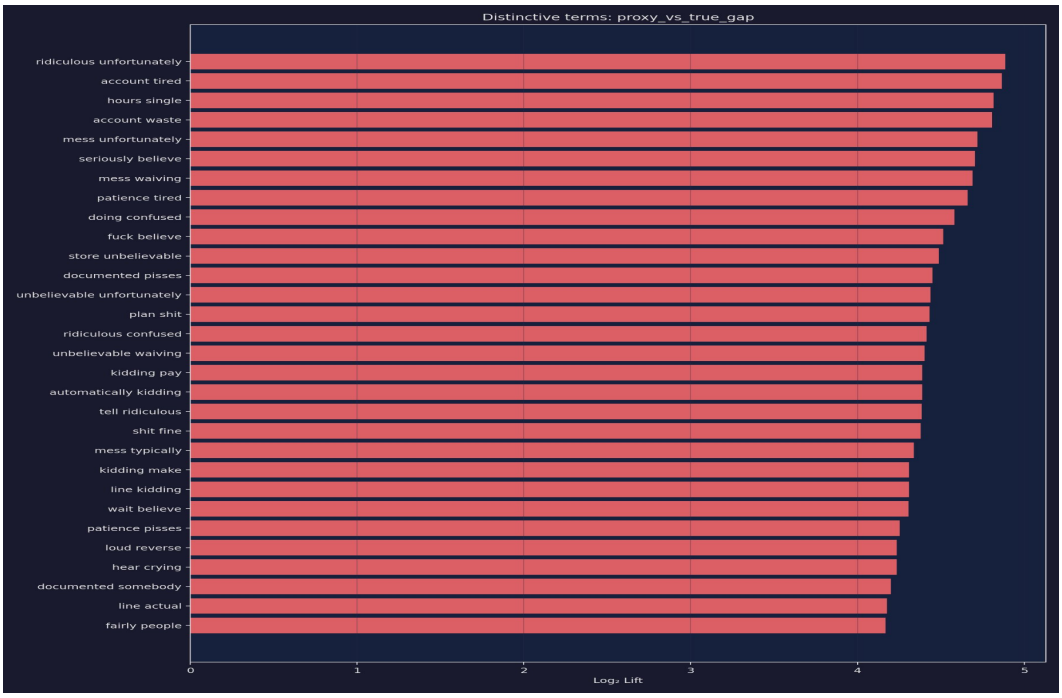

Figure 4 — Distinctive Terms: proxy_vs_true_gap. Sustained negative affect + account references = gap signature.

5. Target-Specific Term Lift: Repeat Contact 30d

Figure 5 shows terms most associated with calls that result in a repeat contact within 30 days. The top terms are 'number goddamn' and 'reference single' (both above 4.0 log2 lift), followed by 'life ticket', 'documented gives', and 'garbage urgent'. The linguistic profile here is distinct from both bandaid and proxy-gap terms: repeat contact calls feature urgency language ('urgent', 'need'), reference to documentation and tickets ('documented gives', 'life ticket', 'reference single'), and expressions of sustained frustration ('goddamn', 'bullshit', 'frustrated').

The mid-range terms include 'error unbelievable', 'months frustrated', 'seriously frustrated', and 'bullshit restarted'—suggesting customers whose issue has persisted over time and who are experiencing the same problem recurring. The presence of 'restarted' and 'cleared losing' suggests technical issues that have been ostensibly fixed but recur. This profile describes the repeat-contact customer: someone whose problem was marked as resolved but who returns because the underlying technical or account issue was never actually addressed.

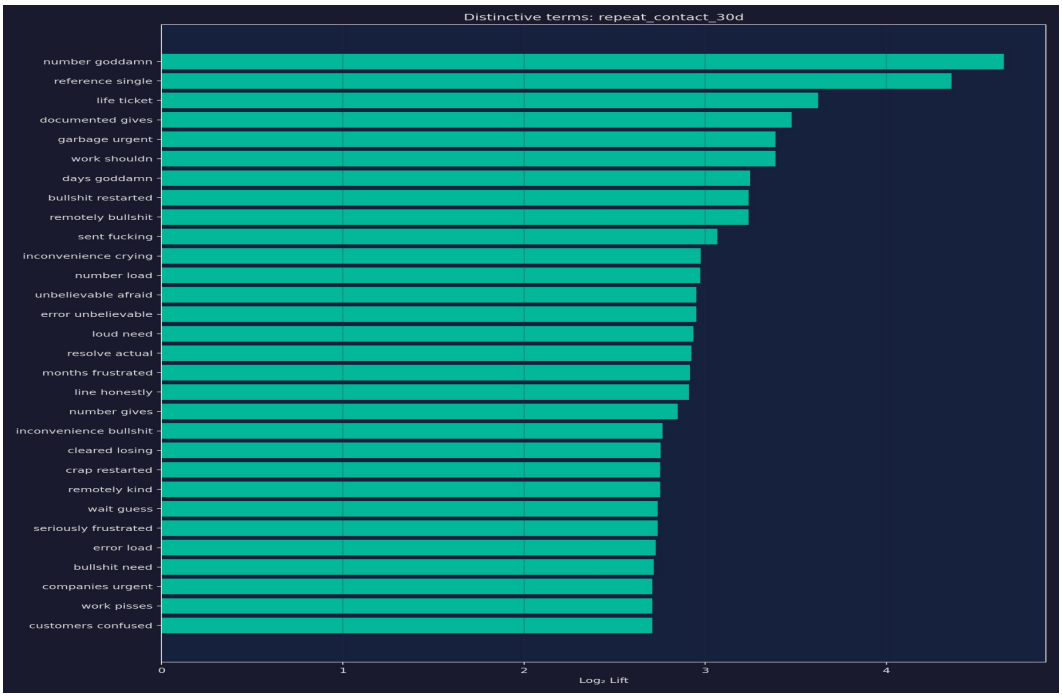

Figure 5 — Distinctive Terms: repeat_contact_30d. Urgency + documentation + recurring issues = repeat contact signature.

6. Conclusion

The transcript analysis reveals three key findings for the NovaWireless governance framework. First, profanity is a viable but blunt predictor of negative outcomes, with 50%+ profanity rates in positive-class calls across all targets. Second, scenario types produce highly distinctive linguistic signatures that could enable automated scenario classification from transcript text alone, without relying on agent-reported scenario labels. Third, each outcome target (bandaid credit, proxy-true gap, repeat contact) has a unique term-lift profile that reflects the underlying mechanism: bandaid calls show forced

positivity over distress, gap calls show sustained negative affect with account references, and repeat calls show urgency with documentation and recurrence language.

These NLP features complement the structured features analyzed in the SII hardening pipeline. The combination of term-lift profiles with the H1 predictive model provides a multi-signal approach to integrity monitoring that operates at the transcript level—detecting mechanism signatures that structured metadata alone cannot capture.